



Development Ticket: TKT-1102

Feature: Secure Cloud Infrastructure Manager (Menu-Driven CLI)

Context

Our company is building a local command-line tool to manage secure cloud servers and databases. The system must store resource details in memory, enforce strict validation, and perform operations like access verification and data encryption. Build this using a Menu-Driven architecture, core OOP principles, Interfaces, Arrays, and String algorithms.



1. The Blueprint (Interface & Abstraction)

1. The Auditable Interface

- Create an interface named `Auditable` with one method: `void generateAuditLog();`

2. The CloudResource Abstract Class

- Create an `abstract` class `CloudResource` that implements `Auditable`.
- Define `private` variables: `resourceId` (String) and `baseCost` (double).
- **Rule for `setResourceId`:** Must start with "RES-" and be exactly 8 characters long. Otherwise, set to "RES-0000".
- **Rule for `setBaseCost`:** Cannot be 0 or negative. Otherwise, set to 10.0.
- Define abstract method: `public abstract double calculateMonthlyCost();`



2. The Models (Inheritance, Polymorphism & Logic)

1. The ComputeServer Class (Child 1)

- Extends `CloudResource`. Variables: `cpuCores` (int), `authorizedUsers` (String Array).
- **Rule for `setCpuCores`:** Must be an EVEN number and positive. Otherwise, set to 2.
- **Rule for `setAuthorizedUsers`:** Array cannot exceed 5 users. If larger, store only the first 5.
- **Overriding:** `calculateMonthlyCost()` returns `baseCost * cpuCores`. `generateAuditLog()` prints ID, cores, and cost.
- **Logic:** Method `boolean verifyUser(String username)` loops through the array. Returns true if found (ignore case).

2. The DatabaseStorage Class (Child 2)

- Extends `CloudResource`. Variable: `storageType` (String).
- **Rule for `setStorageType`:** Must be "SQL" or "NoSQL". Otherwise, set to "UNKNOWN".

- **Overriding:** `calculateMonthlyCost()` returns base cost + 50.0 (if SQL) or + 20.0 (if NoSQL). `generateAuditLog()` prints ID, type, and cost.
- **Logic:** Method `String encryptData(String data)` reverses the string and replaces all vowels (a,e,i,o,u) with '*'. Return the result.



3. The Execution (Menu-Driven Engine & Arrays)

The Main Class

Create a `CloudResource[]` array of size 5 and an `int resourceCount` tracker. Use a `Scanner` and `do-while` loop for the menu:

- **Case 1 (Add Compute Server):** Get inputs, validate via setters, add to array, increment count.
- **Case 2 (Add Database):** Get inputs, validate via setters, add to array, increment count.
- **Case 3 (Verify Access):** Get username, loop array, use `instanceof` to find ComputeServers, call `verifyUser()`.
- **Case 4 (Test Encryption):** Get message, loop array, find a DatabaseStorage via `instanceof`, call `encryptData()` and print.
- **Case 5 (Audit System):** Loop array up to `resourceCount` and call `generateAuditLog()` on each object.
- **Case 6 (Exit):** Break loop.

Note: Ensure your code handles the array reaching its maximum capacity of 5.